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Enrollment No. :- 3250102017

COURSE :- B.C.A. (online) - Semester-2 (2025-26)

SUBJECT :- Mathematices - 2

SUBJECT CODE :- C05BC3201

FACULTY :- Dr. Sunilgar Gourai

ASSIGNMENT :- RUBRICS ASSIGNMENT - 2

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Q. ① :- Determine the absolute and relative errors when $x = 0.00597621$ is rounded to three decimal digits and compare the accuracy with truncation.

Method : 1) BY Rounding.

Ans :→ Given $x = 0.00597621$

$\therefore x = 0.597621 \times 10^{-2}$

$\therefore x_a = 0.598 \times 10^{-2}$ (after rounding off to 3 decimal place)

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→ Absolute Error ; E_A

$\therefore E_A = |x - x_a|$

$\therefore = |0.597621 \times 10^{-2} - 0.598 \times 10^{-2}|$

$\therefore = |-0.000379 \times 10^{-2}|$

$\therefore = 0.000379 \times 10^{-2}$

$\therefore E_A = 0.379 \times 10^{-5}$

→ Relative Error :-

$\therefore E_r = \frac{x - x_a}{x}$

$\therefore = \frac{0.597621 \times 10^{-2} - 0.598 \times 10^{-2}}{0.597621 \times 10^{-2}}$

$\therefore = \frac{0.000379 \times 10^{-2}}{0.597621 \times 10^{-2}}$

$\therefore = \frac{0.379 \times 10^{-5}}{0.597621 \times 10^{-2}}$

$\therefore = 0.634181 \times 10^{-5-(-2)}$

$\therefore E_r = 0.634181 \times 10^{-3}$

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→ Method - 2: By Truncation.

→ ~~Absolute Error;~~ Given $x = 0.00597621$

$\therefore \quad x = 0.597621 \times 10^{-2}$

$\therefore \quad x_a = 0.597 \times 10^{-2}$ (after truncation to three decimal place)

→ Absolute Error;

$\therefore \quad E_A = |x - x_a|$

$\therefore \quad = |0.597621 \times 10^{-2} - 0.597 \times 10^{-2}|$

$\therefore \quad = 0.000621 \times 10^{-2}$

$\therefore \quad E_A = 0.621 \times 10^{-5}$

→ Relative Error;

$\therefore \quad E_r = \dfrac{x - x_a}{x}$

$\therefore \quad = \dfrac{0.597621 \times 10^{-2} - 0.597 \times 10^{-2}}{0.597621 \times 10^{-2}}$

$\therefore \quad = \dfrac{0.621 \times 10^{-5}}{0.597621 \times 10^{-2}}$

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$\therefore$ Er = 1.0391 $\times 10^{-3}$

$\therefore$ Er = 0.10391 $\times 10^{-2}$

Comparison:-

METHOD	Approximate value	Absolute Error	Relative Error
Roundoff	0.598	0.379×10^{-5}	0.634181×10^{-3}
Truncation	0.597	0.621×10^{-5}	0.10391×10^{-2}

Conclusion:-

The round off method gives smaller absolute and relative errors than truncation.

Therfore, rounding is more accurate than truncation for this number.

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Q.(2): Apply the Bisection Method to approximate the root of the equation.

$$x^3 - 4x - 9 = 0$$

in the interval (2.625, 2.75) correct to four significant digits.

Ans:- Bisection Method:-

→ Given, $x^3 - 4x - 9 = 0$

interval (2.625, 2.75)

→ $f(x) = x^3 - 4x - 9$

→ check the interval,

⇒ $f(2.625) = (2.625)^3 - 4(2.625) - 9$

$\therefore \quad = 18.0879 - 10.5 - 9$

$\therefore f(2.625) = -1.4121$

⇒ $f(2.75) = (2.75)^3 - 4(2.75) - 9$

$\therefore \quad = 20.7969 - 11 - 9$

$\therefore f(2.75) = 0.7969$

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Since, $f(2.625) < 0$, $f(2.75) > 0$

There is a root in $(2.625, 2.75)$

→ Bisection Iteration:-

Iteration 1:-

$$x_1 = \frac{a+b}{2} = \frac{2.625 + 2.75}{2}$$

$$= 2.6875$$

Now, $f(2.6875) = (2.6875)^3 - 4(2.6875) - 9$

$$= 19.4109 - 10.75 - 9$$

$$= -0.3391 < 0$$

∴ The root lies in $[2.6875, 2.75]$

Iteration 2:-

→ $x_2 = \frac{2.6875 + 2.75}{2} = 2.71875$

Now, $f(2.71875) = (2.71875)^3 - 4(2.71875) - 9$

$$= 20.0959 - 10.875 - 9$$

∴ $= 0.2209 > 0$

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∴ The root lies in [2.6875, 2.71875]

Iteration 3:-

$$x_3 = \frac{2.6875 + 2.71875}{2} = 2.703125$$

Now, $f(2.703125) = (2.703125)^3 - 4(2.703125) - 9$

$= 19.7514 - 10.8125 - 9$

$= -0.0611 < 0$

∴ The root lies in [2.703125, 2.71875]

Iteration : 4:-

$$x_4 = \frac{2.703125 + 2.71875}{2} = 2.7109375$$

~~Now, $f(2.703125) = (2.703125$~~

Now, $f(2.7109375) = (2.7109375)^3 - 4(2.7109375) - 9$

$= 19.9232 - 10.8438 - 9$

$= 0.0795$

∴ The root lies in [2.703125, 2.7109375]

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→ Iteration : 5 :-

$$x_5 = \frac{2.703125 + 2.7109375}{2} = 2.7070\,3125$$

Now, $f(2.7070\,3125) = (2.7070\,3125)^3 - 4(2.7070\,3125) - 9$

$= 19.837174\,24 - 10.828125 - 9$

$\therefore = 0.0091 > 0$

→ The root lies in $[2.703125, 2.7070\,3125]$

→ Iteration : 7 :-

$$x_6 = \frac{2.703125 + 2.7070\,3125}{2} = 2.7050\,78125$$

Now, $f(2.7050\,78125) = (2.7050\,78125)^3 - 4(2.7050\,78125) - 9$

$\therefore = 19.794\,2676 - 10.820\,3125 - 9$

$\therefore = -0.0260 < 0$

∴ The root lies in $[2.7050\,78125, 2.7070\,3125]$

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→ Iteration: 7:-

$x_7 = \frac{2.7050\,78125 + 2.7070\,3125}{2} = 2.7060\,546875$.

Now, $f(2.7060546875) = (2.7060546875)^3 - 4(2.7060546875) - 9$

$\therefore \quad = 19.8157\,1317 - 10.8242\,1875 - 9$

$\therefore \quad = -0.0084 < 0$

$\therefore$ The root lies in $[2.7060546875, 2.7070\,3125]$

Iteration - 8 :-

$x_8 = \frac{2.7060546875 + 2.7070\,3125}{2} = 2.7065\,4296\,875$

Now, $f(2.7065\,4296\,875) = (2.7065\,4296\,875)^3 - 4(2.7065\,4296\,875) - 9$

$\therefore \quad = 19.8264\,4175 - 10.8261\,7187 - 9$

$\therefore \quad = 0.0003 > 0$

$\therefore$ The root lies in $[2.7060546875, 2.7065\,4296\,875]$

→ Approximate Root: $x = 2.706$ $[2.7061, 2.7065]$

Thus, the value of the root of the equation $x^3 - 4x - 9 = 0$ correct upto 4 decimal place

$x = 2.706$.

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Q. (8): Analyze the dataset and implement Newton's Backward Interpolation Formula to estimate $y(4.25)$, and interpret the accuracy of the result.

x	2.5	3.0	3.5	4.0	4.5
y	9.75	12.45	15.70	19.52	23.75

Ans:— ✓ Backward Difference Table:-

x_i	y_i	∇	∇^2	∇^3	∇^4
$x_0 = 2.5$	$y_0 = 9.75$				
		2.7			
$x_1 = 3.0$	$y_1 = 12.45$		0.55		
		3.25		0.02	
$x_2 = 3.5$	$y_2 = 15.70$		0.57		−0.18
		3.82		−0.16	
$x_3 = 4.0$	$y_3 = 19.52$		0.31		
		4.23			
$x_4 = 4.5$	$y_4 = 23.75$				

— ✓ Newton's Backward Difference formula is,

$$y(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!}\nabla^4 y_n + \dots$$

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Where, $p = \frac{x - x_n}{h}$

→ $h = 0.5$

Also, $x = 4.25$

Now, $p = \frac{x - x_4}{h} = \frac{4.25 - 4.5}{0.5} = \frac{-0.25}{0.5} = -0.5$

$$y(4.25) = y_4 + p\nabla y_4 + p(p+1)\frac{\nabla^2 y_4}{L^2} + p(p+1)(p+2)\frac{\nabla^2 y_4}{L^3}$$
$$+ p(p+1)(p+2)(p+3)\frac{\nabla^4 y_4}{24}$$

→ $y(4.25) = 23.75 + (-0.5)(4.23) + (-0.5)(-0.5+1)\frac{0.41}{2} +$
$(-0.5)(-0.5+1)(-0.5+2)\left(-\frac{0.16}{6}\right) + (-0.5)$
$(-0.5+1)(-0.5+2)(-0.5+3)\left(-\frac{0.18}{24}\right)$

$\therefore y(4.25) = 23.75 - 2.115 - 0.05125 + 0.0100 + 0.0070$

$\therefore y(4.25) = 21.6008$

Accuracy Interpretation:

In Newton's Backward Interpolation, the final Interpretation

$y(4.25) \approx 21.601$

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is reliable and accurate interpolation because:

(1) The data is equally spaced.

(2) Higher-order differences are small.

(3) And the interpolation point lies close to the end of the table where Newton's Backward formula performs best.

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Q.(4) Evaluate the first derivative $f'(0.50)$ using suitable numerical differentiation techniques.

x	0.50	0.75	1.00	1.25	1.50
$y=f(x)$	0.13	0.42	1.00	1.95	2.35

Ans:- Newton's forward differences Differentiation :-

$$y'(x_0) = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} + \frac{\Delta^4 y_0}{4} + \dots\right]$$

Here, $x_0 = 0.5$, $x_1 = 0.75$, $x_2 = 1.00$, $x_3 = 1.25$ And $x_4 = 1.50$

Also, $y_0 = 0.13$, $y_1 = 0.42$, $y_2 = 1.00$, $y_3 = 1.95$ & $y_4 = 2.35$

Again $h = 0.75 - 0.50 = 0.25$

Consider the following forward difference table for the given data.

x_i	y_i	Δy_i	$\Delta^2 y_i$	$\Delta^3 y_i$	$\Delta^4 y_i$
0.50	0.13				
		0.29			
0.75	0.42		0.29		
		0.58		0.08	
1.00	1.00		0.37		-1.00
		0.95		-0.92	
1.25	1.95		-0.55		
		0.40			
1.50	2.35				

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from the table,

$\Delta y_0 = 0.29$, $\Delta^2 y_0 = 0.29$ $\Delta^3 y_0 = 0.08$ And $\Delta^4 y_0 = -1.00$

Therefor, by Newton's forward differentiation formula.

$$y'(x_0) = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4}\right]$$

$$\therefore y'(0.50) = \frac{1}{0.25}\left[0.29 - \frac{0.29}{2} + \frac{0.08}{3} - \frac{1.00}{4}\right]$$

$$\therefore = \frac{1}{0.25}\left[0.29 - 0.145 + 0.027 + 0.25\right]$$

$$\therefore = \frac{1}{0.25}\left[0.422\right]$$

$$\therefore = 1.688$$

$$\therefore \boxed{y'(0.50) = 1.688}$$

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Q.5: Evaluate the estimation of $f'(6)$ using numerical differentiation.

x	0	1	2	3	4	5	6
$y = f(x)$	0	2.5	8.5	15.5	24.5	36.5	50

Ans:— Newton's backward differentiation:

x_i	y_i	∇y_i	$\nabla^2 y_i$	$\nabla^3 y_i$
0	0			
		2.5		
1	2.5		3.5	
		6		-2.5
2	8.5		1	
		7		1
3	15.5		2	
		9		1
4	24.5		3	
		12		-1.5
5	36.5		1.5	
		13.5		
6	50			

→ Using Newton's backward differentiation formula:-

$$f'(x_n) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \dots\right]$$

→ Substitute the values at $x = 6$, stepsize $h = 1$.

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$$\therefore f'(6) = \frac{1}{1}\left[13.5 + \frac{1}{2}(9.5) + \frac{1}{3}(-1.5)\right]$$

$$\therefore = 13.5 + 0.75 - 0.5$$

$$\therefore \boxed{f'(6) = 13.75}$$